

AT30K Telnet Connection Guide

Prerequisites

Information You Need

- **IP Address** of the AT30K device
- **System ID** (1-254, configured in the device)
- **Password** (if restricted access is needed): `3179636424`
- **Port** (typically 23 for Telnet, but check device documentation)

Network Requirements

- Network connectivity to the AT30K device
- Telnet client (Windows Command Prompt, PuTTY, or terminal)

Step-by-Step Connection Process

Step 1: Open Telnet Connection

```
bash
```

```
# Windows Command Prompt
```

```
telnet [IP_ADDRESS] [PORT]
```

```
# Example
```

```
telnet 192.168.1.100 23
```

```
# Linux/Mac Terminal
```

```
telnet 192.168.1.100 23
```

Step 2: Initial Communication Test

Once connected, send an ESC character to clear the command buffer:

Method 1: ESC Key

- Press the `ESC` key on your keyboard

Method 2: Hex Code

- Type: `1B` (hex code for ESC)

Expected Response (if communication port is open):

```
<CR><LF>  
MON 30A>
```

Expected Response (if in listen-only mode):

```
No immediate response - device is in listen-only mode
```

Step 3: Establish Communication

Enter the address debug command with your system ID:

```
AD,XX
```

Where **XX** is your system ID in hex (01-FE)

Examples:

- System ID 1: **AD,01**
- System ID 4: **AD,04**
- System ID 16: **AD,10**
- System ID 255: **AD,FF** (this puts device in listen-only mode)

Expected Response:

```
<CR><LF>  
MON 30A>
```

Step 4: Enable Restricted Access (if needed)

For advanced configuration commands:

```
PW,3179636424
```

Expected Response:

```
pw,XXXXXXXXXX ENABLED<CR><LF>  
MON 30A>
```

Basic Commands to Test Connection

Get System Information

bash

Get TDR firmware version

PV

Get Communication firmware version

CV

Get USB firmware version

UV

Get cable status

CS

Get System Configuration

bash

Get system ID

DR,20FD,20FD

Get IP address

DR,2100,2103

Get Language setting

DR,20E9,20E9

Connection Troubleshooting

Problem: No Response After Telnet Connection

Solutions:

1. Verify IP address and port
2. Check network connectivity: `ping [IP_ADDRESS]`
3. Ensure device is powered on
4. Try different port numbers (23, 502, 10001)

Problem: Connection Refused

Solutions:

1. Check if Telnet is enabled on the device
2. Verify firewall settings
3. Ensure device is not in a fault state
4. Try connecting via serial port instead

Problem: Connected but No Command Response**Solutions:**

1. Send ESC character first: Press `ESC` key or type `1B`
2. Try different system IDs (01, 02, 03, etc.)
3. Check if device is in listen-only mode
4. Verify correct ASCII protocol settings

Problem: "Address Debug" Fails**Solutions:**

1. Verify correct system ID in hex format
2. Check EEPROM system ID: `DR, 20FD, 20FD`
3. Try system ID range 01-FE
4. Ensure device is not locked by another session

Sample Session

```
C:\> telnet 192.168.1.100 23
```

```
Connected to 192.168.1.100
```

```
[Press ESC key]
```

```
MON 30A>AD,01
```

```
MON 30A>PW,3179636424
```

```
pw,3179636424 ENABLED
```

```
MON 30A>CS
```

```
12000003
```

```
21000003
```

```
32000005
```

```
42000006
```

```
MON 30A>DR,20FD,20FD
```

```
20FD 01
```

```
MON 30A>AD,FF
```

```
ad,ff
```

Important Notes

Command Format

- Commands are **NOT case sensitive**
- Use comma separators where specified
- Hex values can be upper or lowercase
- Always end editing sessions with proper cleanup

Session Management

- Use `AD,FF` to close communication (listen-only mode)
- Use `WD` command after configuration changes (requires reconnection)
- Only one active session typically allowed

Safety Considerations

- **Backup configurations** before making changes

- **Test commands** on non-critical systems first
- **Document changes** for maintenance records
- **Coordinate with operations** before accessing live systems

Data Retrieval Examples

```
bash
```

```
# Get cable 1 setup data
```

```
DP,1,1,6000,61FF
```

```
# Get current map for cable 1
```

```
DP,1,1,4040,4FFF
```

```
# Get reference map 1 for cable 1
```

```
DP,1,1,6218,69D7
```

```
# Get archive pointer
```

```
AP
```

```
# Get Last 4 archives
```

```
AR,XXXX
```

Advanced Configuration

Network Settings Retrieval

```
bash
```

```
# IP Address
```

```
DR,2100,2103
```

```
# Subnet Mask
```

```
DR,2104,2107
```

```
# Gateway
```

```
DR,2108,210B
```

```
# MAC Address (Last 3 bytes)
```

```
DR,2120,2122
```

System Status Checks

```
bash
```

```
# Monitor flag status
```

```
DR,20F0,20F0
```

```
# Autosilence setting
```

```
DR,20EA,20EA
```

```
# Serial port settings
```

```
DR,20F1,20F2
```

```
# Relay behavior
```

```
DR,2137,2137
```

Remember to always close your session properly with **AD,FF** when finished to allow other users to connect.